UNIVERSITY OF UYO

DEPARTMENT OF MATHEMATICS

MTH 314 **COMPLEX ANALYSIS I**

Instructions: Answer only four questions ***TIME:*** 2 HOURS

1. Define in terms of polar coordinate (x,y) and variables x and y we define them as for each pair of real number x and y,z real complex number.

   Define an outstanding mathematic at achievement by the theorem of Gauss.

   Interpret in terms of polar coordinate the coordinates system.

2(a) Given the equation

$$z = x + iy$$

and carefully define the real and the imaginary parts. Define the angle θ and hence define $z = \theta = \tan . {}^{y}\!/_{x}$.

(b) Given $i^2 = -1$ and $z = |z|^2$

then compute $(z_1 + z_2^{\frac{y}{2}})$

$$z_1 \cdot z_2$$

$$\frac{z_1}{z_2}$$

3(a) Write z_1 z_2 as in polar form and compute z_1 z_2 and hence compute

$$\arg(z_1 z_2) = \arg(z_1) + \arg z_2 .$$

(b) What set of points in the x-y complex plane satisfies

$$\left| \frac{z}{z-1} \right| = 2$$

4(a) Express the complex numbers $3 + 6i$ in polar form and also divide it by $2 - 3i$.

(b) Find the three cube roots of unity of $z = i$

Find the three roots of $z \quad -1$

Find the four solutions of $z^4 + 16 = 6$

5(a) Arguments of $z = -1 + i$ and $z_2 = 1 - i$ so that $\arg z_1 - z_2$.

(b) Find z_1 and z_2 so that $0 \le z_1 \arg z_2 < 2\pi$.

$(-1+i) - (1-i)$

$-1+i-1+i$

$-2+2i$

$(-2, 2i)$